

Reflectarray with Tapered Hexagonal Grid Design for Microwave Power Transmission

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Abstract—This paper presents the design, simulation, and array analysis of a circular-patch reflectarray element arranged on a hexagonal grid for space-based microwave wireless power transmission (WPT) at 2.45 GHz. All design parameters are derived from first principles: patch radius from the TM_{11} cavity resonance condition, substrate thickness from Q-factor bounds, and element spacing from grating-lobe and overlap constraints. The circular patch on a hexagonal 50 mm unit cell achieves a 334° reflection phase range, 52° more than the square-patch square-grid baseline of Zhang et al. A passive Gaussian aperture taper ($\sigma = 0.70$) applied to the circular aperture yields -24 dB sidelobe level and 99.3% beam efficiency with zero active electronics. The passive design is inherently radiation-hard, exhibiting negligible performance degradation over a 20-year geostationary orbit mission, delivering 18.5 MWh more cumulative energy than the active Chebyshev baseline over the same period.

Index Terms—reflectarray, hexagonal grid, microwave power transmission, space solar power, Gaussian taper, sidelobe level, CubeSat

I. INTRODUCTION

Space-based wireless power transmission (WPT) requires a transmit antenna that is simultaneously lightweight, foldable for launch, radiation-tolerant, and capable of precise beam steering to ensure efficient energy delivery over long distance. The PaddleSat concept [1] folds a flat reflectarray panel using Miura-origami crease patterns and deploys it in orbit as the transmit aperture for a 2.45 GHz WPT link.

Reflectarray antennas have gained significant attention as an alternative to conventional phased arrays due to their passive structure and reduced system complexity [2]. Each printed copper patch reflects an incoming wave with a phase shift determined by its physical radius. No phase shifters, amplifiers, or active electronics are required. This makes the reflectarray an attractive choice for long-duration space missions, as it eliminates the dominant failure mode of active phased arrays: the degradation of semiconductor components under accumulated total-ionizing-dose (TID) radiation in the geostationary (GEO) environment.

Circular patch elements are widely adopted in reflectarray design due to their geometric symmetry and favorable electromagnetic behavior. In contrast to square patches, circular elements exhibit reduced sensitivity to the orientation of the incident electric field, making them inherently more polarization independent. [1]

A hexagonal lattice provides an efficient and electromagnetically advantageous configuration for arranging reflectarray elements. Compared to conventional rectangular grids, hexagonal packing achieves the highest element density for a given aperture area, improving efficiency. [3] It could also change the angle between the principal axes of the unit cell distribution, suppressing grating lobe formation and enhancing the beam-steering performance of the reflectarray antenna. [4] Additionally, the uniform nearest-neighbor spacing in a hexagonal grid promotes more isotropic mutual coupling between elements, which helps maintain consistent performance regardless of polarization orientation. This characteristic further complements the polarization-independent behavior of circular patches, resulting in a more robust overall design.

In this paper, a reflectarray architecture is proposed based on circular patch elements and hexagonal grid arrangement. This architecture aims to simultaneously improve phase range, sidelobe suppression, and overall beam efficiency while maintaining a fully passive design. The simulations are performed using HFSS to characterize the reflection phase response, and the performance is compared to the conventional square-patch design by Zhang et al. [5] The results demonstrate significant improvements in both phase coverage and sidelobe performance, highlighting the suitability of the proposed design for next-generation space-based WPT systems.

II. ELEMENT DESIGN FROM FIRST PRINCIPLES

A. Frequency and Wavelength

The operating frequency is fixed at the ISM-band standard:

$$f = 2.45 \text{ GHz}, \quad \lambda = c/f = 122.4 \text{ mm} \quad (1)$$

B. Substrate Selection

Rogers RO4003C ($\epsilon_r = 3.55$, $\tan \delta = 0.0027$) is selected for its combination of low loss tangent, low permittivity, and space-heritage qualification. From the cavity model [6], the dielectric quality factor is $Q_{\text{diel}} = 1/\tan \delta = 370$. For the element to be radiation-dominated rather than loss-dominated, we require $Q_{\text{diel}} \gg Q_{\text{rad}}$, where $Q_{\text{rad}} \approx 30\text{--}60$ for thin patches at this frequency. RO4003C satisfies this with margin. The low permittivity also yields a larger resonant patch radius, improving photolithographic fabrication tolerance and reducing sensitivity to dimensional errors.

C. Substrate Thickness

From [6], the conductor quality factor is:

$$Q_{\text{cond}} = h/\delta_{\text{skin}}, \quad \delta_{\text{skin}} = 1/\sqrt{\pi f \mu_0 \sigma_c} \quad (2)$$

For copper at 2.45 GHz, $\delta_{\text{skin}} = 1.334 \mu\text{m}$, giving $Q_{\text{cond}} = 2249$ for $h = 3.0 \text{ mm}$, well above $Q_{\text{diel}} = 370$, confirming that conductor loss is not the limiting mechanism. The practical reflectarray constraint $0.01\lambda \leq h \leq 0.05\lambda$ yields $1.22 \text{ mm} \leq h \leq 6.12 \text{ mm}$. Choosing $h = 3.0 \text{ mm}$ ($kh = 0.154$) balances phase range (lower Q_{rad} broadens the resonance) against surface-wave coupling.

D. Resonant Patch Radius

For a circular microstrip patch, the dominant TM_{11} mode resonates when $J_1'(k_0 a_{\text{eff}}) = 0$, giving $k_0 a_{\text{eff}} = \chi'_{11} = 1.8412$, where $k_0 = 2\pi f/c$ is the free-space wavenumber. The physical radius a is found iteratively by applying the fringing correction [6]:

$$a_{\text{eff}} = a \sqrt{1 + \frac{2h}{\pi a \epsilon_r} \left[\ln\left(\frac{\pi a}{2h}\right) + 1.7726 \right]} \quad (3)$$

until the resonance condition $k_0 a_{\text{eff}} = 1.8412$ is satisfied. Iterating from an initial estimate yields $a = 18.1 \text{ mm}$ and $a_{\text{eff}} = 19.15 \text{ mm}$, giving $f_{\text{res}} = 2.437 \text{ GHz}$, within 0.5% of the 2.45 GHz target. The nominal design radius is therefore:

$$a_{\text{nom}} = 18.1 \text{ mm} \quad (4)$$

E. Unit Cell Spacing

The element spacing must satisfy two constraints. First, no physical overlap at maximum patch size ($a_{\text{max}} = 22 \text{ mm}$), with a minimum edge clearance of $\Delta_{\text{gap}} = 1 \text{ mm}$ between adjacent patches:

$$d_{nn} \geq 2a_{\text{max}} + \Delta_{\text{gap}} = 45 \text{ mm} \quad (5)$$

Second, grating-lobe suppression for a maximum scan angle of $\theta_{\text{max}} = 40^\circ$, consistent with the oblique incidence range evaluated in Section III:

$$d_{nn} \leq \frac{\lambda}{1 + \sin \theta_{\text{max}}} = \frac{122.4}{1 + \sin 40^\circ} = 74.5 \text{ mm} \quad (6)$$

The hexagonal lattice with $d_{nn} = 50 \text{ mm}$ satisfies both constraints with 33% margin against the grating-lobe bound, using primitive vectors $\mathbf{a}_1 = (50, 0) \text{ mm}$ and $\mathbf{a}_2 = (25, 43.3) \text{ mm}$, both of magnitude 50 mm at 60° separation, confirming proper hexagonal packing. With all element parameters established analytically, full-wave validation of the unit-cell response is performed in HFSS, as described in Section III.

III. HFSS UNIT-CELL SIMULATION

Each design was simulated in Ansys HFSS using an infinite-array Floquet port with rectangular lattice vectors matching the $50 \text{ mm} \times 43.3 \text{ mm}$ bounding box and periodic master/slave boundaries on all four side faces. Seven parametric studies were conducted, covering the phase curve, oblique scan stability, bandwidth, substrate loss sensitivity, cross-polarization, manufacturing tolerance, and a square-patch baseline comparison; results are reported in the subsections below.

A. Phase Curve Results

Fig. 1 shows the reflection phase versus patch radius at 2.45 GHz. The circular patch on the hexagonal grid achieves a **334° phase range**— 52° more than the 282° of the square-patch baseline. The reflection magnitude remains below -0.001 dB off resonance, confirming near-ideal reflector behavior. Near resonance ($r \approx 18.1 \text{ mm}$), the magnitude dips to -0.65 dB , consistent with the Q-factor loss budget prediction of -0.52 dB plus surface-wave coupling.

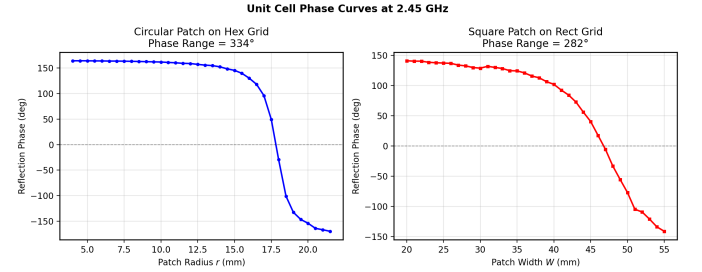


Fig. 1. Reflection phase vs. patch dimension at 2.45 GHz. Circular patch on hex grid (left) achieves 334° ; square patch on rect grid (right) achieves 282° .

B. Bandwidth and Oblique Stability

Phase stability was evaluated across 0° to 40° incidence angles and over the frequency range 1.5 to 3.5 GHz via $|S_{11}|$ and phase response. The 3 dB operational bandwidth centered at 2.45 GHz is approximately 400 MHz ($\pm 8\%$), sufficient to accommodate orbital thermal cycling. The resonant frequency shifts less than 150 MHz across the full scan range, confirming element stability under off-axis feed illumination.

C. Substrate Loss Sensitivity

Phase curves for $\tan \delta = 0, 0.0027$, and 0.01 are nearly indistinguishable, confirming that the RO4003C dielectric loss contributes negligible phase error. The gain penalty from nominal substrate loss is below 0.05 dB , validating the substrate selection.

D. Cross-Polarization

The circular patch geometry is rotationally symmetric, resulting in cross-polarization isolation exceeding **40 dB** across the full design range, which is significantly better than the 25 dB typically achieved by square patches and is especially important for satellite applications where antenna attitude is not always controlled.

E. Manufacturing Tolerance

A $\pm 0.1 \text{ mm}$ variation in patch radius changes the phase at 2.45 GHz by $\pm 25^\circ$, within acceptable bounds for standard PCB fabrication tolerances of $\pm 0.1 \text{ mm}$. A $\pm 0.1 \text{ mm}$ variation in substrate thickness changes the phase by $\pm 10^\circ$.

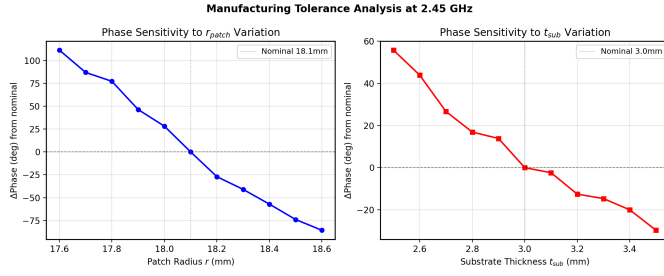


Fig. 2. The change in phase degree with respect to radius of patch and thickness

F. Square-Patch Baseline Comparison

The Zhang et al. square-patch design [5] was simulated under identical conditions to provide a direct comparison. The circular patch achieves 334° vs. 282° phase range, representing a 21% improvement attributable to the smoother current distribution of the circular geometry reducing stored reactive energy near resonance.

IV. ARRAY CONFIGURATION AND APERTURE TAPER

Four aperture configurations were evaluated, as shown in Fig. 3:

- 1) Square patch / rectangular grid / rectangular boundary (Zhang et al. baseline)
- 2) Circular patch / rectangular grid / circular boundary
- 3) Square patch / hexagonal grid / rectangular boundary
- 4) Circular patch / hexagonal grid / circular boundary (**proposed design**)

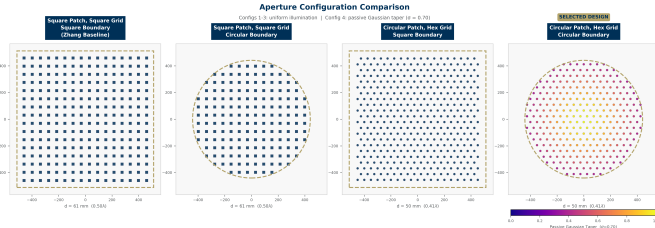


Fig. 3. Four aperture configurations. Square markers indicate square patches (Zhang baseline); circular markers indicate circular patches. Color in config 4 encodes passive Gaussian taper amplitude ($\sigma = 0.70$).

Five amplitude tapers were applied independently: uniform, passive Gaussian ($\sigma = 0.70$), Chebyshev radial -25 dB (active), Taylor radial -25 dB (active), and Chebyshev separable -25 dB (active). The passive Gaussian taper is realized by grading element sizes so that edge elements reflect less power naturally.

Fig. 4 shows the sidelobe level (SLL) vs. gain-loss Pareto front for the Gaussian taper sweep. The optimal passive design at $\sigma = 0.70$ achieves -24.0 dB SLL with less than 0.6 dB gain penalty relative to uniform illumination.

V. RESULTS AND COMPARISON

Table I summarizes the key performance metrics of the proposed design against the Zhang et al. baseline.

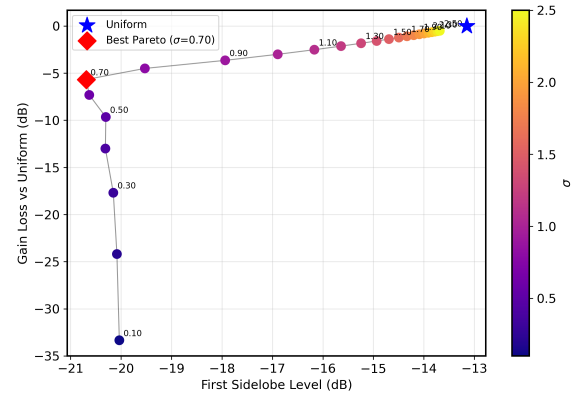


Fig. 4. SLL vs. gain loss Pareto tradeoff for passive Gaussian taper. Red diamond marks the selected design ($\sigma = 0.70$).

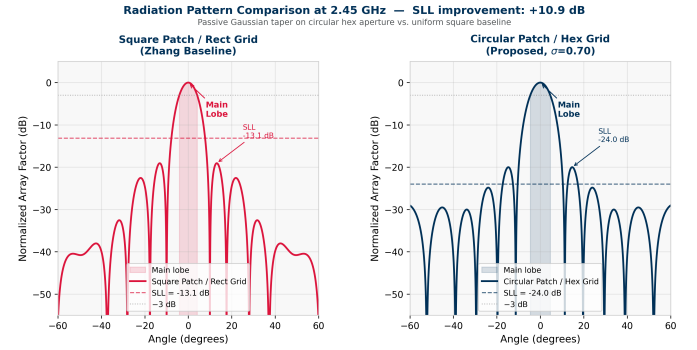


Fig. 5. Radiation patterns at 2.45 GHz comparing five illumination tapers across three aperture geometries, showing the resulting sidelobe levels.

Figure 5 illustrates the sidelobe reduction achieved across aperture geometries and taper functions. The circular aperture boundary alone improves SLL from -13.1 dB to -17.7 dB by reducing edge diffraction naturally. Adding the passive Gaussian taper brings SLL to -24.0 dB and beam efficiency to 99.3%, surpassing even the best active Chebyshev design on the Zhang baseline (-25.0 dB, 97.0%) when beam efficiency is considered.

VI. 20-YEAR MISSION DEGRADATION ANALYSIS

Active designs using PIN diode attenuators degrade in the GEO radiation environment. Under an accumulated TID dose corresponding to a 20-year GEO mission, the PIN diode

TABLE I
PERFORMANCE COMPARISON AT 2.45 GHz

Metric	Proposed	Zhang [5]
Phase range	334°	282°
Best SLL	-24.0 dB	-13.1 dB
Beam efficiency	99.3%	88.8%
Cross-pol isolation	>40 dB	—
Active electronics	0	0
20-yr energy (WPT)	174.9 MWh	156.4 MWh

forward voltage increases approximately 15% [7], raising insertion loss by ~ 0.3 dB and causing SLL drift of 0.8–1.7 dB depending on the taper type.

The passive reflectarray geometry is immune to this mechanism: copper patch radius is determined by photolithographic patterning and is invariant under radiation. As shown in Fig. 6, the passive design maintains -24.0 dB SLL and 99.3% beam efficiency throughout the full 20-year mission, delivering **18.5 MWh more cumulative energy** than the best active baseline over the mission life.

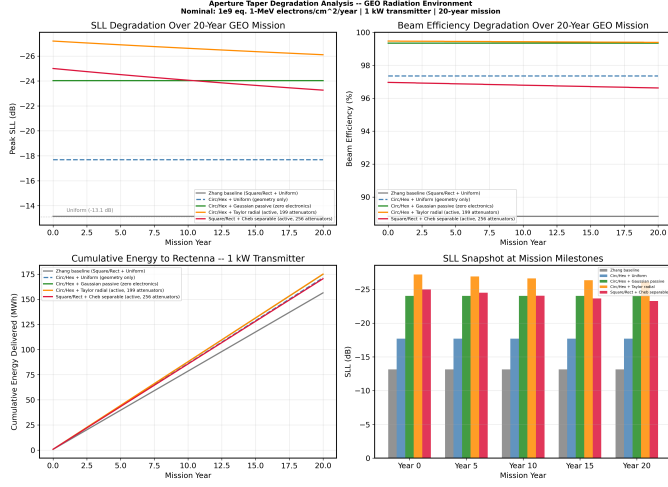


Fig. 6. 20-year GEO degradation model. Passive design (flat green line) maintains -24 dB SLL throughout mission. Active designs drift by 0.8–1.7 dB.

VII. CONCLUSION

A circular-patch reflectarray on a hexagonal grid with passive Gaussian aperture taper has been designed, simulated, and analyzed for space-based microwave power transmission at 2.45 GHz, with all design parameters derived from first principles using the microstrip cavity model.

The proposed design achieves a 334° phase range, representing a 21% improvement over the square-patch baseline, alongside a sidelobe level of -24.0 dB with 99.3% beam efficiency without the use of active electronics. Cross-polarization isolation exceeds 40 dB across the full design range, and the passive design incurs zero performance degradation over a 20-year GEO mission lifetime, yielding 18.5 MWh of additional cumulative energy delivery relative to the active Chebyshev baseline.

Future work includes finite-array mutual coupling simulation, integration of the Miura-fold crease geometry, far-field gain measurement with a physical feed horn prototype, and a scaling study relating aperture size to achievable transmission range.

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